Welcome to the

AI Agent Lab

a Google Cloud Hackathon by Devoteam

Oslo - March 11th

devoteam

Google Cloud

Google Cloud
DIAMOND
Services Partner

AI Agent Lab

a Google Cloud Hackathon by Devoteam

Opening Remarks

Thomas Bruun
Sales Manager
Devoteam

devoteam

Google Cloud

Google Cloud
DIAMOND
Services Partner

Agenda

09:00 - 09:15	**Welcome & Opening Remarks** {Speaker name, job title at Devoteam}
09:15 - 09:45	**ADK Technical enablement** Review your agentic concept and the business value it will bring.
09:45 - 12:30	**Hacking | Phase 1** Initial development phase with on-site support from Google Cloud and Devoteam AI experts.
12:30 - 13:30	**Lunch** Refuel with a buffet-style selection of food and refreshments.
13:30 - 16:30	**Hacking | Phase 2** Final development sprint. Focus on refining your agentic solution and polishing your presentation.
16:30 - 17:30	**Showcase & Judging** Each team will present their agentic solution and share the business value to our panel of Google AI experts, with a prize awarded to the winning team.
17:30 - 19:30	**Networking & Drinks** Celebrate the day with a selection of refreshments while building connections with fellow Agentic AI innovators.

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Tech Onboarding

ADK (Agent Development Kit)

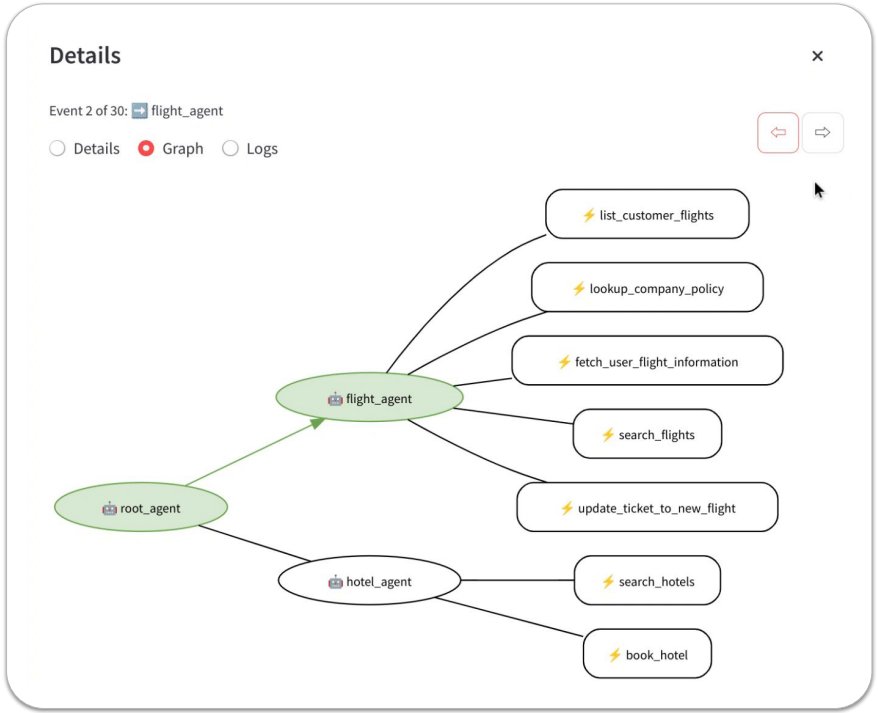
Marco Pellegrino
ML/AI Engineer

What is ADK?

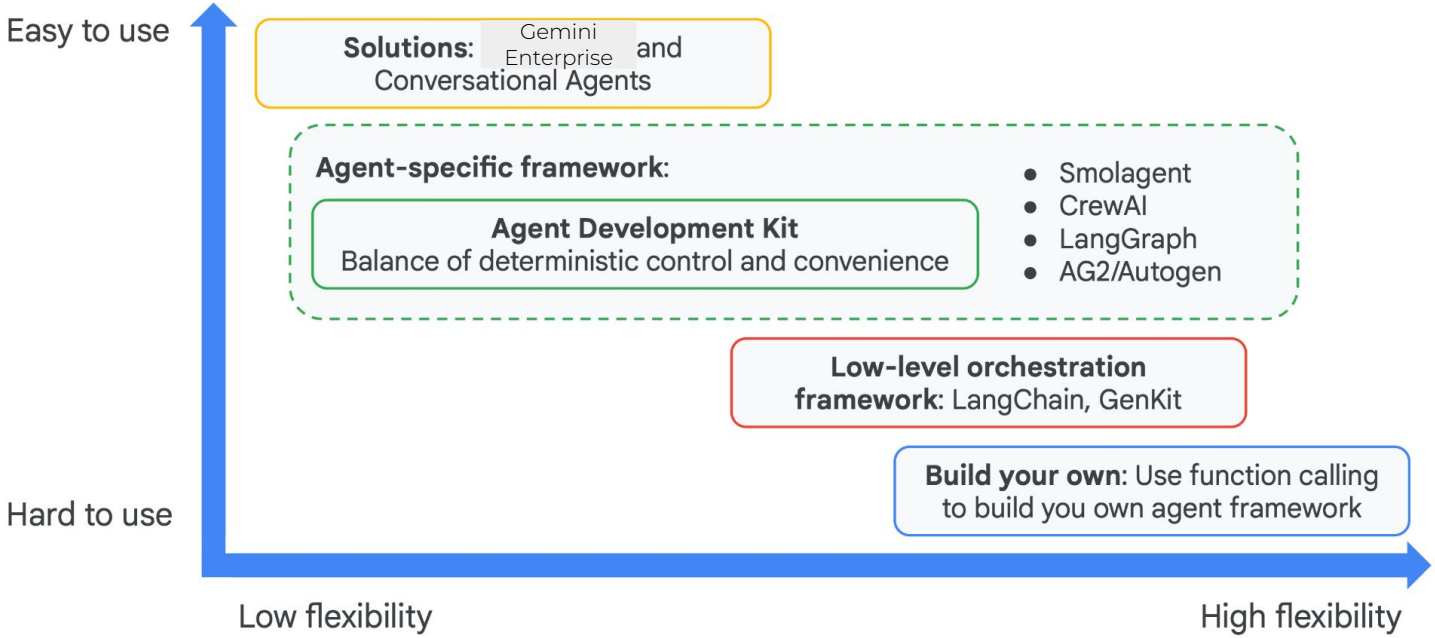
Agent Development Kit

Framework and SDK to build multi-agent solutions

- **Python**, **Java**, **Javascript**, and **Go** SDKs
- Define **multi-agent** applications orchestrating actions across many agents and tools
- Interleave **deterministic** logic with actions driven by **gen AI reasoning** for effortless, hybrid agents
- Gemini is the default, but **any generative AI model** is supported, including fine-tuned models



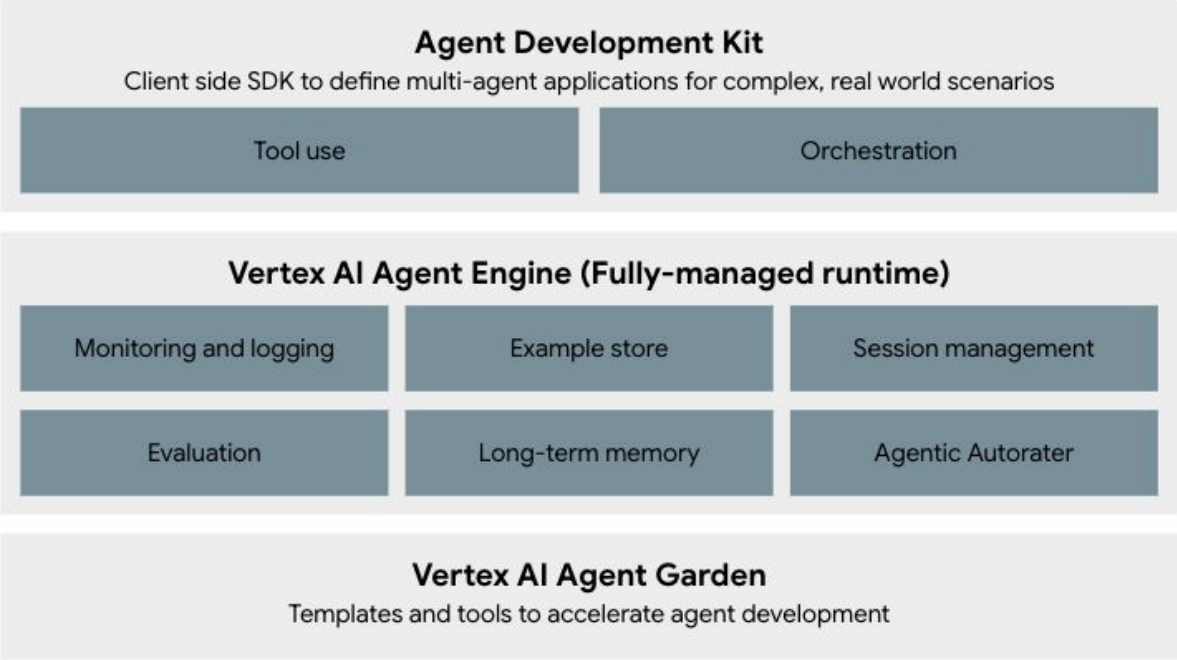
Higher abstraction makes system easier to build



What does ADK include ?

- Multi-agent systems
- Prebuilt, custom, and community tools
- Support for evaluation
- A convenient local development UI with debugging tools
- Callbacks to invoke functions throughout the flow
- Session memory for stateful conversations
- Collaboration on documents with shared artifacts
- Can deploy to Agent Engine for fully-managed agent infrastructure

Discover, Build and Deploy within GCP



Boilerplate code to create an Agent

```python
root_agent = Agent(
    name="greeting_agent",
    model="gemini-2.5-pro",
    description="Greeting agent",
    instruction="""
    You are a helpful assistant that greets the user.
    Ask for the user's name and greet them by name.
    """
)
```

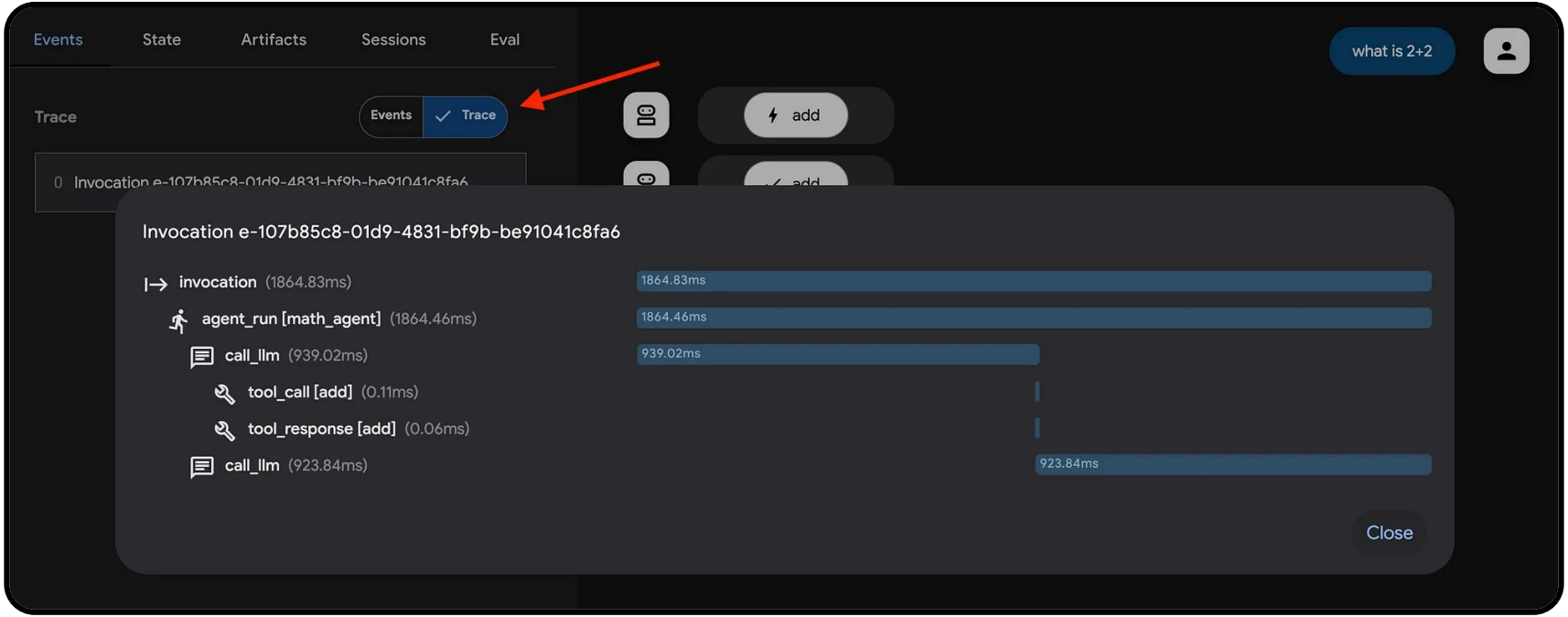
Built-in Tools

```
root_agent = Agent(
    name="tool_agent",
    model="gemini-2.5-pro",
    description="Tool agent",
    instruction="""
    You are a helpful assistant that can use the following tools:
    - google_search
    """,
    tools=[google_search]
)
```

Custom Tools

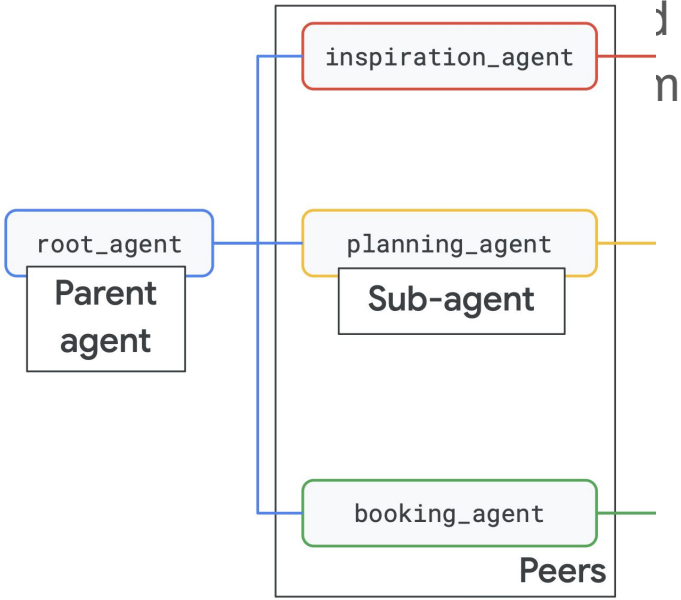
```python
def get_current_time() -> dict:
    """
    Get the current time in the format YYYY-MM-DD HH:MM:SS
    """
    return {
        "current_time": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
    }


root_agent = Agent(
    name="tool_agent",
    model="gemini-2.5-pro",
    description="Time agent",
    instruction="""
    You are a helpful assistant that can use get the current time using the following tools:
    - get_current_time
    """,
    tools=[get_current_time]
)
```

Tracing


Events
State
Artifacts
Sessions
Eval
what is 2+2
Trace
Events
Trace
add
0 Invocation e-107b85c8-01d9-4831-bf9b-be91041c8fa6
Invocation e-107b85c8-01d9-4831-bf9b-be91041c8fa6
invocation (1864.83ms)
1864.83ms
agent_run [math_agent] (1864.46ms)
1864.46ms
call_llm (939.02ms)
939.02ms
tool_call [add] (0.11ms)
tool_response [add] (0.06ms)
call_llm (923.84ms)
923.84ms
Close

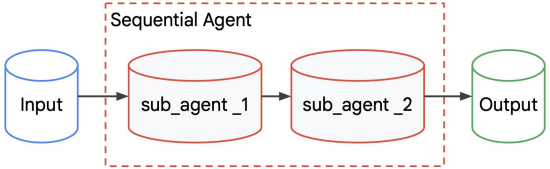
Multi-Agents

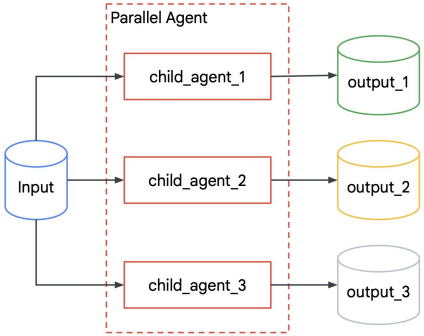
- Instead of one AI, it's a system where multiple agents work together to achieve a goal.
- Complex problem d to specialized agents.
- For example, one night write code, and a third could review the f

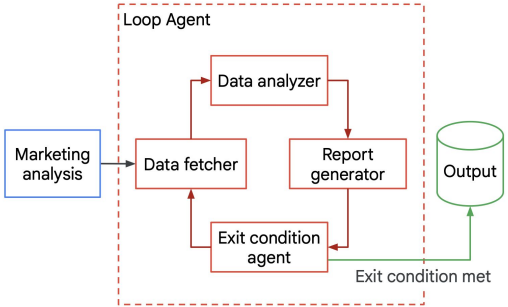


Workflow Agents

- Create a controlled flow directly from agent to agent - Deterministic
- The LLMs don't make decisions compared to those setup via LLM-based model agents
- Different types:
  - Sequential Agent
  - Loop Agent

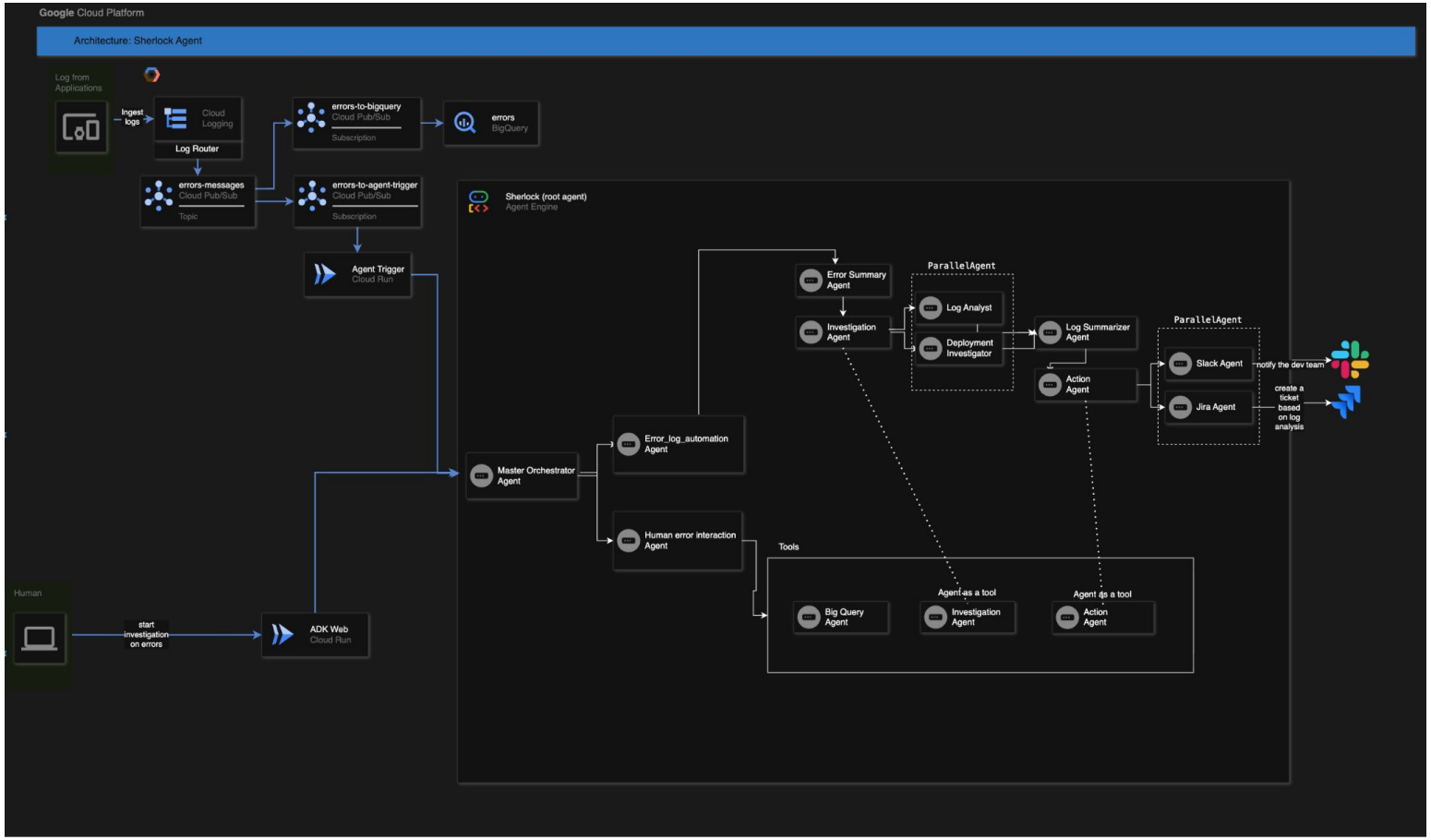






Sherlock Agent

- **Problem Statement:** Manual daily review of error logs, which is time-consuming and risks overlooking critical issues.
- **Solution:** Implement an agent to automate log checking, issue detection, version rollback, and notification (Jira ticket creation and Slack message).


Google Cloud Platform
Architecture: Sherlock Agent
Log from Applications
Ingest logs
Cloud Logging
Log Router
errors-messages
Cloud Pub/Sub
Topic
errors-to-bigquery
Cloud Pub/Sub
Subscription
errors
BigQuery
errors-to-agent-trigger
Cloud Pub/Sub
Subscription
Agent Trigger
Cloud Run
Sherlock (root agent)
Agent Engine
Error Summary Agent
Investigation Agent
ParallelAgent
Log Analyst
Deployment Investigator
Log Summarizer Agent
Action Agent
ParallelAgent
Slack Agent
Jira Agent
notify the dev team
create a ticket based on log analysis
Master Orchestrator Agent
Error_log_automation Agent
Human error interaction Agent
Tools
Big Query Agent
Agent as a tool
Investigation Agent
Agent as a tool
Action Agent
Human
start investigation on errors
ADK Web
Cloud Run

The possibilities are endless!

How do we start?

```
pip install google-adk
```

Ways to test your agent

- Command Line Interface

```
adk run my_google_search_agent # Replace with your agent's folder name
```

- Web UI

```
adk web
```

Set up Google Cloud CLI for Google Cloud access

Home > Documentation > Application development > Google Cloud SDK > Guides

Was this helpful?

Install the Google Cloud CLI

Send feedback

This quickstart describes the recommended method to install and initialize the Google Cloud CLI. After initialization, run a few core gcloud CLI commands to view information about your installation and verify it was successful.

To follow step-by-step guidance for this task directly in the Google Cloud console, click **Guide me**:

Guide me

Install ADK

```
pip install google-adk
```

Set environment variables

Ensure your environment is configured for Vertex AI:

1. **Authentication:** Use Application Default Credentials (ADC):

```
gcloud auth application-default login
```

2. **Environment Variables:** Set your project and location:

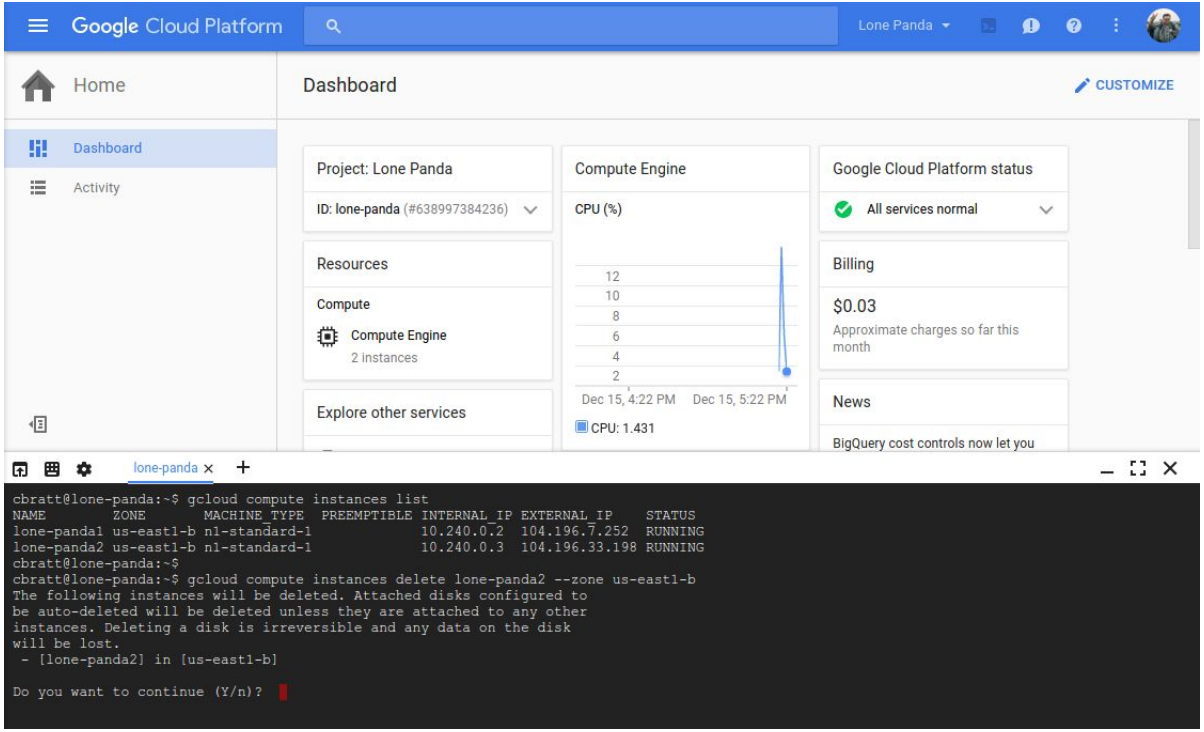
```
export GOOGLE_CLOUD_PROJECT="YOUR_PROJECT_ID"
export GOOGLE_CLOUD_LOCATION="YOUR_VERTEX_AI_LOCATION" # e.g., us-cent
```

3. **Enable Vertex Backend:** Crucially, ensure the `google-genai` library targets Vertex AI:

```
export GOOGLE_GENAI_USE_VERTEXAI=TRUE
```

What if I don’t want to install on my machine?

Cloud Shell: a machine on the cloud!



My suggestion for today

1) Build you agent locally. Have fun with prompts, flows, and AI
2) Play with Google Cloud deployments

🚀 Agent Starter Pack

pypi v0.39.0 | 1-Minute Overview | Documentation | Try in Firebase Studio | Launch in Cloud Shell | Stars 5.9k

A Python package that provides **production-ready templates** for GenAI agents on Google Cloud.

Focus on your agent logic—the starter pack provides everything else: infrastructure, CI/CD, observability, and security.

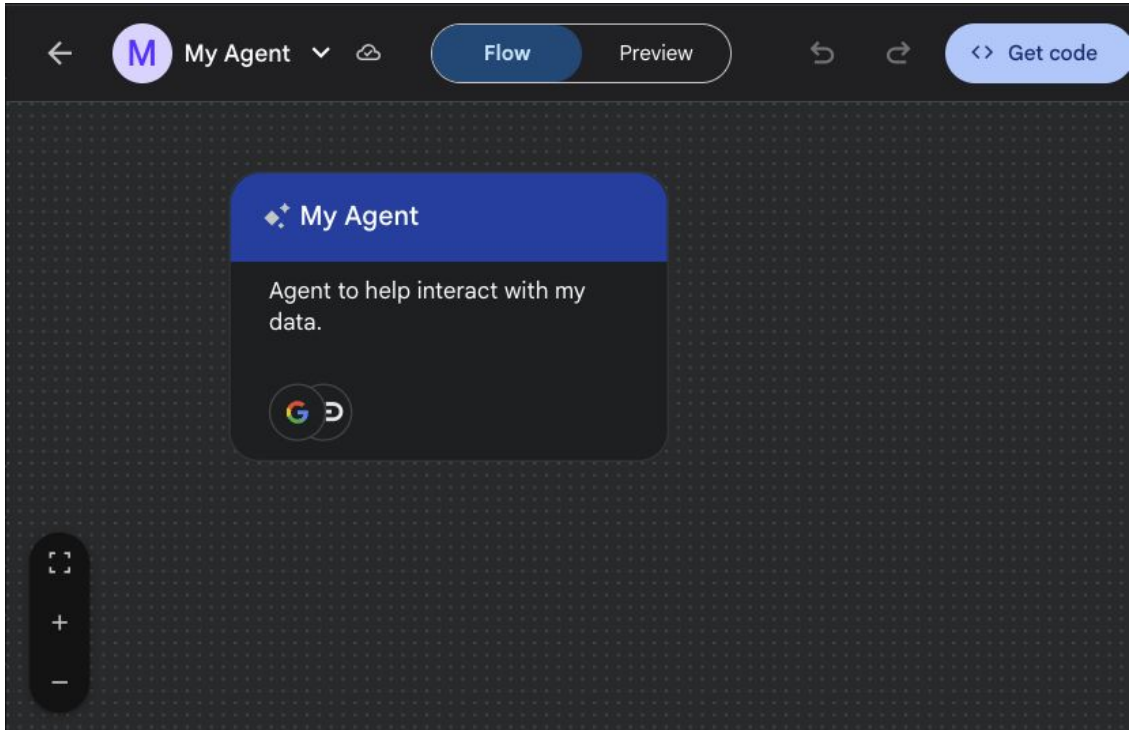
⚡ Launch	🧪 Experiment	✅ Deploy	🛠️ Customize
Pre-built agent templates (ReAct, RAG, multi-agent, Live API).	Vertex AI evaluation and an interactive playground.	Production-ready infra with monitoring, observability, and CI/CD on Cloud Run or Agent Engine.	Extend and customize templates according to your needs. 🆕 Now integrating with Gemini CLI

3) Build real connectors: the tough part! Not recommended for today

What if I don’t know how to code?

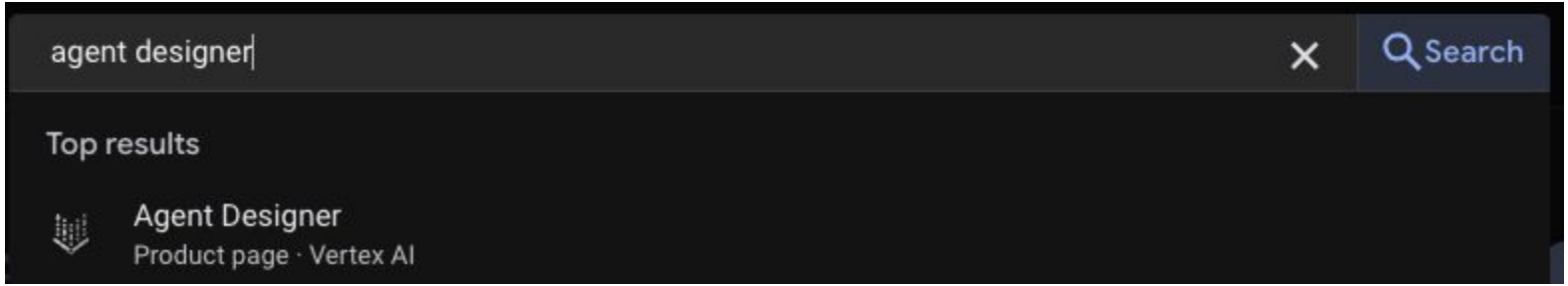
Google Cloud Agent Designer

No-code agentic framework



Google Cloud Agent Designer

How do I find it?



I don’t have GCP

Reach-out to me and I’ll provide you a demo account.
Remember to export your data after the hackathon

Any questions?

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Final Ideation

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Let's **Hack!**

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Lunch & Networking

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Showcase & Judging

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Closing Remarks

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Thank you!
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Let's **Network!**

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